

Regular rings and perfectoid towers

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September 10, 2026

The 12th Japan-Vietnam Joint Seminar on Commutative Algebra
at Khanh Hoa University, Vietnam

This talk is based on the following:

- K. Hayashi, *Regular rings and perfectoid towers*, [arXiv:2605.27337](#).

Contents

1. Regular rings and perfect(oid) algebras
2. Regular rings and perfectoid towers

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Kunz's theorem¹

For a Noetherian $\mathbb{F}_p$ -algebra R , TFAE.

- ① R is **regular**.
 - ② The absolute Frobenius $\varphi: R \rightarrow R$ is (faithfully) flat.
 - ③ The perfect R -algebra $R_{\text{perf}} := \varinjlim \left(R \xrightarrow{\varphi} R \xrightarrow{\varphi} \cdots \right)$ is faithfully flat.
 - ④ There exists a **perfect R -algebra** A that is faithfully flat.
- An $\mathbb{F}_p$ -algebra A is **perfect** $\stackrel{\text{def}}{\iff} \varphi: A \rightarrow A$ is bijective.
 - For ④ $\Rightarrow$ ②, observe the following commutative diagram:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & R \\ \text{faithfully flat} \downarrow & & \downarrow \text{(faithfully) flat} \\ A & \xrightarrow[\varphi]{\cong} & A \end{array}$$

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Perfectoid algebras are generalizations of perfect $\mathbb{F}_p$ -alg. to mixed char.

p -adic Kunz's theorem (B. Bhatt–S. B. Iyengar–L. Ma²)

For a Noetherian ring R such that $p \in \text{rad}(R)$, TFAE.

- ① R is **regular**.
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e.g., $R = \mathbb{Z}_p \rightarrow A = \widehat{\mathbb{Z}_p[p^{\frac{1}{p^\infty}}]} = \varinjlim \left(\mathbb{Z}_p \hookrightarrow \mathbb{Z}_p[p^{\frac{1}{p}}] \hookrightarrow \mathbb{Z}_p[p^{\frac{1}{p^2}}] \hookrightarrow \cdots \right)^\wedge$.

A ring A is **perfectoid** (w.r.t. a fixed prime p) $\iff \exists \varpi \in A$ s.t.

- ① $p \in \varpi^p A$, and A is ϖ -adically complete.

②

$$\begin{array}{ccc} A/\varpi^p A & \xrightarrow{\varphi} & A/\varpi^p A \\ \downarrow & & \parallel \\ A/\varpi A & \xrightarrow{\cong} & A/\varpi^p A \end{array}$$

- ③ $A_{\varpi\text{-tor}} \xrightarrow{\sim} A_{\varpi\text{-tor}}; x \mapsto x^p$. Here $A_{\varpi\text{-tor}} := \bigcup_{n>0} (0 :_A \varpi^n)$.

²Comm. Algebra **47**(6), (2019), 2367–2383.

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Perfectoid towers are introduced to axiomatize towers such as $\mathbb{Z}_p \hookrightarrow \mathbb{Z}_p[p^{1/p}] \hookrightarrow \cdots \hookrightarrow \mathbb{Z}_p[p^{1/p^i}] \hookrightarrow \cdots$.

Definition (S. Ishiro–K. Shimomoto–K. Nakazato³)

An inductive system $\{R_i\}_{i \geq 0} = (R_0 \rightarrow R_1 \rightarrow \cdots \rightarrow R_i \xrightarrow{t_i} \cdots)$ of Noetherian rings indexed by $\mathbb{N}$ is a (Noetherian) **perfectoid tower** (arising from (R_0, p)) $\stackrel{\text{def}}{\iff} \exists \varpi \in R_1$ s.t.

① $pR_1 = \varpi^p R_1$, and $p \in \text{rad}(R_i)$.

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$$\begin{array}{ccc} R_{i+1}/pR_{i+1} & \xrightarrow{\varphi} & R_{i+1}/pR_{i+1} \\ \downarrow & \searrow \textcolor{blue}{\text{\scriptsize $\exists F_i$}} & \uparrow \overline{t_i} \\ R_{i+1}/\varpi R_{i+1} & \dashrightarrow^{\cong} & R_i/pR_i \end{array}$$

③ $p(R_i)_{p\text{-tor}} = 0$, and F_i induces $(R_{i+1})_{p\text{-tor}} \xrightarrow{\sim} (R_i)_{p\text{-tor}}$.

$\implies \left(\varinjlim_i R_i \right)_p^\wedge$ is a perfectoid ring.

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To a perfectoid tower $\{R_i\}_{i \geq 0}$, we associate the perfectoid tower of $\mathbb{F}_p$ -algebras $\{R_i^{s.b}\}_{i \geq 0}$ (called the **tilt** of $\{R_i\}_{i \geq 0}$) defined by

$$R_i^{s.b} \stackrel{\text{def}}{=} \varprojlim \left(R_i/pR_i \xleftarrow{F_i} R_{i+1}/pR_{i+1} \xleftarrow{F_{i+1}} R_{i+2}/pR_{i+2} \xleftarrow{F_{i+2}} \cdots \right).$$

Examples

- ① $\{R_i\}_{i \geq 0} = \{\mathbb{Z}_p[p^{1/p^i}]\}_{i \geq 0} \rightsquigarrow \{R_i^{s.b}\}_{i \geq 0} = \{\mathbb{F}_p[[T]][T^{1/p^i}]\}_{i \geq 0}$.
- ② A perfectoid tower of $\mathbb{F}_p$ -algebras is precisely a **perfect tower**, i.e., is isomorphic to $(R \xrightarrow{\varphi} R \xrightarrow{\varphi} R \xrightarrow{\varphi} \cdots)$ for some Noetherian $\mathbb{F}_p$ -alg. R .
- ③ R. Ishizuka and Ishiro–Shimomoto constructed from:
 - affine semigroup rings $\mathbb{Z}_p[[H]]$ ($H \subset \mathbb{N}^{\oplus n}$).

$$\begin{aligned} \mathbb{Z}_p[[t^2, t^3]] &\hookrightarrow \mathbb{Z}_p[p^{1/p}][[t^{2/p}, t^{3/p}]] \hookrightarrow \mathbb{Z}_p[p^{1/p^2}][[t^{2/p^2}, t^{3/p^2}]] \hookrightarrow \cdots, \\ \mathbb{F}_p[[T]][[t^2, t^3]] &\hookrightarrow \mathbb{F}_p[[T^{1/p}][[t^{2/p}, t^{3/p}]] \hookrightarrow \mathbb{F}_p[[T^{1/p^2}][[t^{2/p^2}, t^{3/p^2}]] \hookrightarrow \cdots. \end{aligned}$$

- p -Stanley–Reisner rings $\mathbb{Z}_p[[x_2, \dots, x_n]]/(\square \text{ free monomials in } p, x_2, \dots, x_n)$.

$$\begin{aligned} \mathbb{Z}_p[[x, y]]/(px, py) &\hookrightarrow \mathbb{Z}_p[p^{1/p}][[x^{1/p}, y^{1/p}]]/(p^{1/p}x^{1/p}, p^{1/p}y^{1/p}) \hookrightarrow \cdots \\ \mathbb{F}_p[[T, x, y]]/(Tx, Ty) &\hookrightarrow \mathbb{F}_p[[T^{1/p}, x^{1/p}, y^{1/p}]]/(T^{1/p}x^{1/p}, T^{1/p}y^{1/p}) \hookrightarrow \cdots \end{aligned}$$

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Theorem A

$\{R_i\}_{i \geq 0}$: a perfectoid tower ($\rightsquigarrow \{R_i^{s,b}\}_{i \geq 0}$). Fix $i \geq 0$. Then TFAE.

- ① R_i is **regular**.
- ② $t_i: R_i \rightarrow R_{i+1}$ is (faithfully) flat.
- ③ $R_i^{s,b}$ ($\cong R_0^{s,b}$) is **regular**.
- ④ $t_i^{s,b}: R_i^{s,b} \rightarrow R_{i+1}^{s,b}$ ($\cong \varphi: R_0^{s,b} \rightarrow R_0^{s,b}$) is (faithfully) flat.

In particular, both ① and ② are independent of i .

Corollary B (Rigidity of regularity)

R_i is regular for **some** $i \implies R_i$ is regular for **any** $i \geq 0$.

Corollary C (Kunz-type theorem in terms of perfectoid towers)

For a Noetherian local ring $(R, \mathfrak{m})$ such that $p \in \mathfrak{m}$, TFAE.

- ① R is **regular**.
- ② There exists a **perfectoid tower starting from R** whose i -th transition map t_i is (faithfully) flat for any **(or, equivalently, some)** $i \geq 0$.

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Corollary C gives another proof of the following result without the regularity criterion of Auslander–Buchsbaum–Serre.

Corollary D

If $(R, \mathfrak{m})$ is a regular local ring such that $p \in \mathfrak{m}$, then $R_{\mathfrak{p}}$ is regular for every $\mathfrak{p} \in V(pR)$.

- Ishizuka–Nakazato⁴ also gave another proof using [prismatic Kunz's theorem](#). While their proof relies on the methods from [homotopy theory](#), our proof is based on a more elementary argument.

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Theorem A (recall)

Fix $i \geq 0$. Then TFAE.

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Sketch of proof We may assume that R_i is local.

③ $\Leftrightarrow$ ④: Kunz's theorem.

① $\Rightarrow$ ②: Show " R_{i+1} is CM" & " $\dim(R_{i+1}) = \dim(R_i)$."

② $\Leftrightarrow$ ④: Use the following results of Ishiro–Nakazato–Shimomoto:

- $R_i^{s,b}/p^{s,b}R_i^{s,b} \xrightarrow{\cong} R_i/pR_i$ (e.g., $\mathbb{F}_p[[T]]/(T) \xrightarrow{\cong} \mathbb{F}_p = \mathbb{Z}_p/p\mathbb{Z}_p$).
- p is R_i -regular $\iff p^{s,b}$ is $R_i^{s,b}$ -regular. $\dots (*)$

The case $(*)$ follows from the **local criterion of flatness**. Reduce the general case to $(*)$ via $\widetilde{R}_i := R_i/(R_i)_{p\text{-tor}} (\rightsquigarrow (\widetilde{R}_i)^{s,b} \cong R_i^{s,b}/(R_i^{s,b})_{p^{s,b}\text{-tor}})$.

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Sketch of proof We may assume that R_i is local.

③ $\Leftrightarrow$ ④: Kunz's theorem.

① $\Rightarrow$ ②: Show " R_{i+1} is CM" & " $\dim(R_{i+1}) = \dim(R_i)$."

② $\Leftrightarrow$ ④: Use the following results of Ishiro–Nakazato–Shimomoto:

- $R_i^{s,b}/p^{s,b}R_i^{s,b} \xrightarrow{\cong} R_i/pR_i$ (e.g., $\mathbb{F}_p[[T]]/(T) \xrightarrow{\cong} \mathbb{F}_p = \mathbb{Z}_p/p\mathbb{Z}_p$).
- p is R_i -regular $\iff p^{s,b}$ is $R_i^{s,b}$ -regular. $\dots (*)$

The case $(*)$ follows from the **local criterion of flatness**. Reduce the general case to $(*)$ via $\widetilde{R}_i := R_i/(R_i)_{p\text{-tor}} (\rightsquigarrow (\widetilde{R}_i)^{s,b} \cong R_i^{s,b}/(R_i^{s,b})_{p^{s,b}\text{-tor}})$.

Theorem A (recall)

Fix $i \geq 0$. Then TFAE.

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② + ③ $\Rightarrow$ ①: Suggested by O. Gabber, J. Lurie⁵ proved the following.

A mixed characteristic analogue of Kunz's theorem

$(R, \mathfrak{m})$: Noetherian local. $\varpi \in \mathfrak{m}$: R -regular with $p \in \varpi^p R$. TFAE.

- ① R is **regular**.
- ② The homomorphism $\psi: R/\varpi R \rightarrow R/\varpi^p R; x \mapsto x^p$ is (faithfully) flat.

③ $\Rightarrow p^{s.b}$ is $R_i^{s.b}$ -regular $\iff p$ is R_i -regular $\iff \varpi$ is R_{i+1} -regular.

Hence one can apply Theorem above to R_{i+1} . Observe:

$$\begin{array}{ccc}
 R_{i+1}/pR_{i+1} & \xrightarrow{\varphi} & R_{i+1}/pR_{i+1} \\
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Thus R_{i+1} is regular, and since t_i is faithfully flat, R_i is regular. $\square$

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Summary

- ① (Kunz '69) TFAE for $R \supset \mathbb{F}_p$.
 - R is regular.
 - $\varphi: R \rightarrow R$ is (faithfully) flat.
 - There exists a **perfect R -algebra** A that is faithfully flat.
- ② (Bhatt–lyengar–Ma '19) TFAE for R with $p \in \text{rad}(R)$.
 - R is regular.
 - There exists a **perfectoid R -algebra** A that is faithfully flat.
- ③ (Gabber–Lurie '23) TFAE for $(R, \mathfrak{m})$ with $\varpi \in \text{NZD}(R)$ & $p \in \varpi^p R$.
 - R is regular.
 - $R/\varpi R \rightarrow R/\varpi^p R; x \mapsto x^p$ is (faithfully) flat.
- ④ (H. '26) TFAE for $(R, \mathfrak{m})$ with $p \in \mathfrak{m}$.
 - R is regular.
 - There exists a **perfectoid tower starting from R** whose i -th transition map is (faithfully) flat for any (or, equivalently, some) $i \geq 0$.